

CUSUM Change-Point Detection Analysis

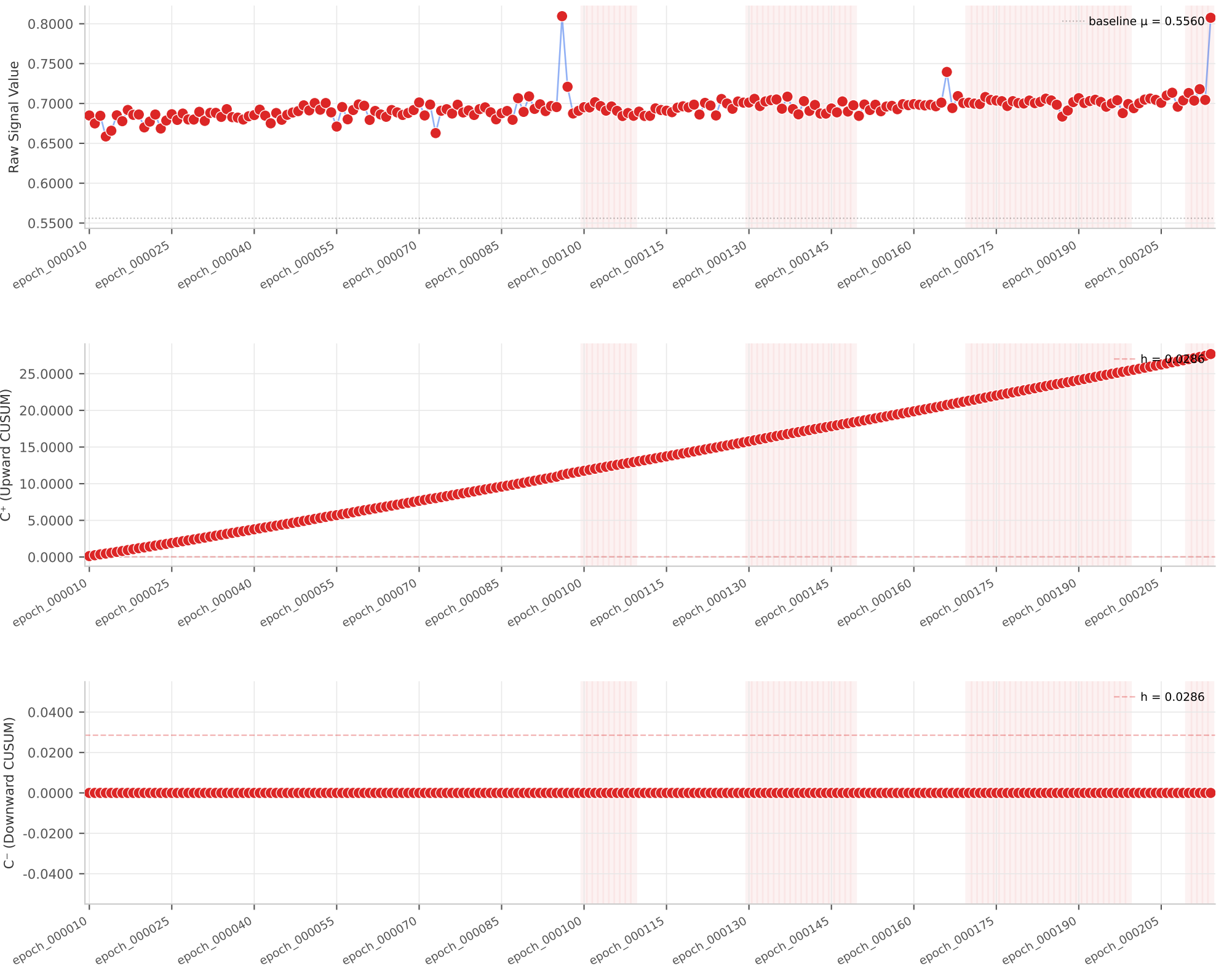
Cumulative Sum Control Chart | JS Divergence + New IP Ratio + Top-K Source

Run Parameters		Baseline CUSUM Parameters		Detection Summary (Any Signal)	
Dataset	txtD280_5.0_1.0_cusum_sparse.json	Baseline JS	$\mu=0.5560$ $\sigma=0.0071$ $k=0.0036$ $h=0.0286$	True Positives (TP)	65
Timestamp	2026-02-23T14:48:38.244345	New IP Ratio	$\mu=0.0000$ $\sigma=0.0000$ $k=0.0000$ $h=0.0000$	True Negatives (TN)	140
Method	CUSUM Change-Point (JS + New IP + TopK)	Top-K	$\mu=0.0622$ $\sigma=0.0031$ $k=0.0015$ $h=0.0123$	False Positives (FP)	0
Total epochs	—	False Negatives (FN)			0
Baseline epochs	— (?%)	Precision			0.317
Attack epochs	205	Recall			1.000
Malicious traffic	?%	F1 Score			0.481
CUSUM k-factor	0.5	Detection Rate			100.0%
CUSUM h-factor	4.0				
Mode	SPARSE				
Attack probability	0.3				
Window size	10				
Windows attacked	7				

Combined JS Divergence

CUSUM Signal Analysis | malicious = 0% | baseline = 5%

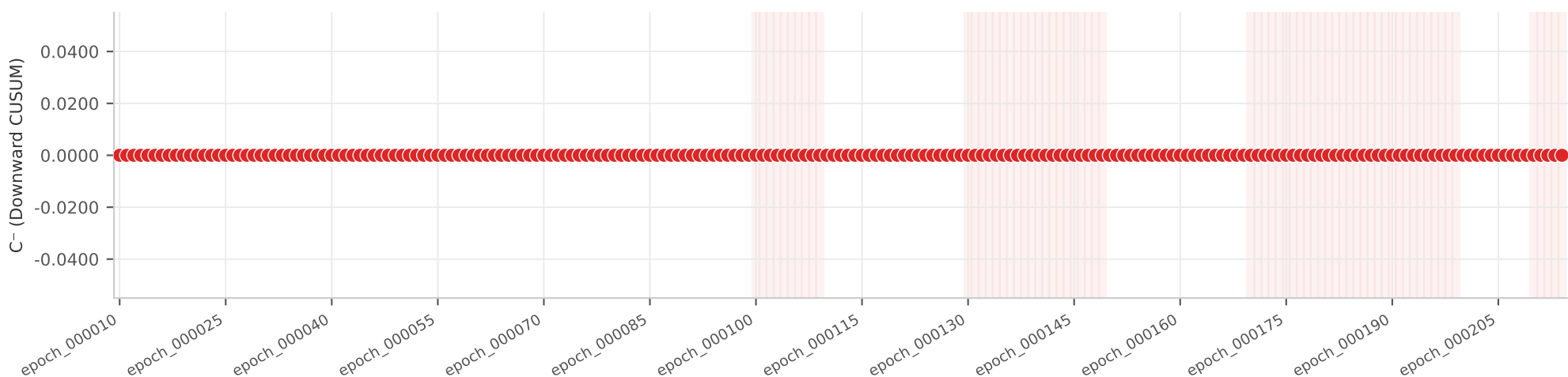
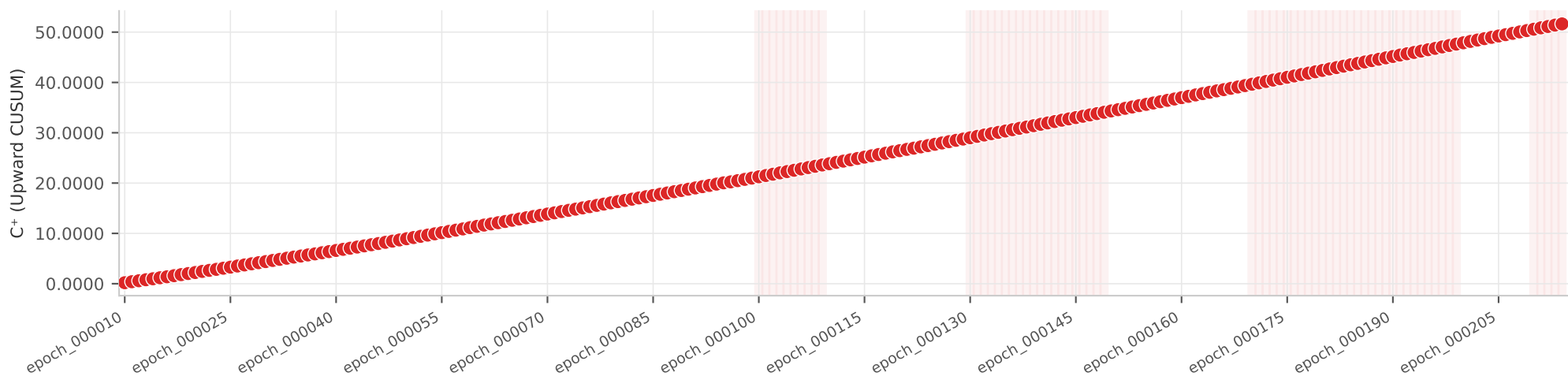
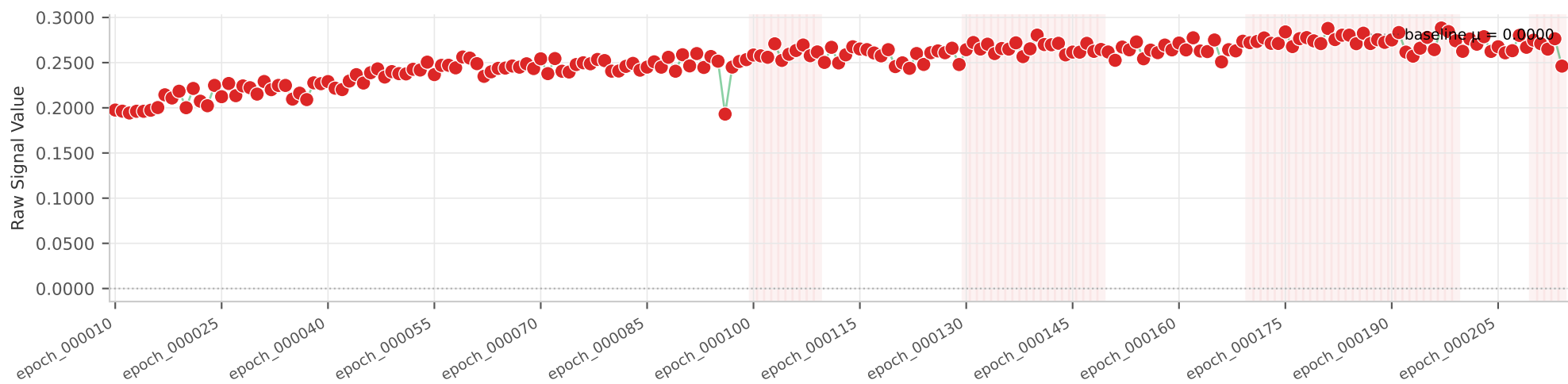
- CUSUM alarm fired
- Attacked epoch



New Source IP Ratio

CUSUM Signal Analysis | malicious = 0% | baseline = 5%

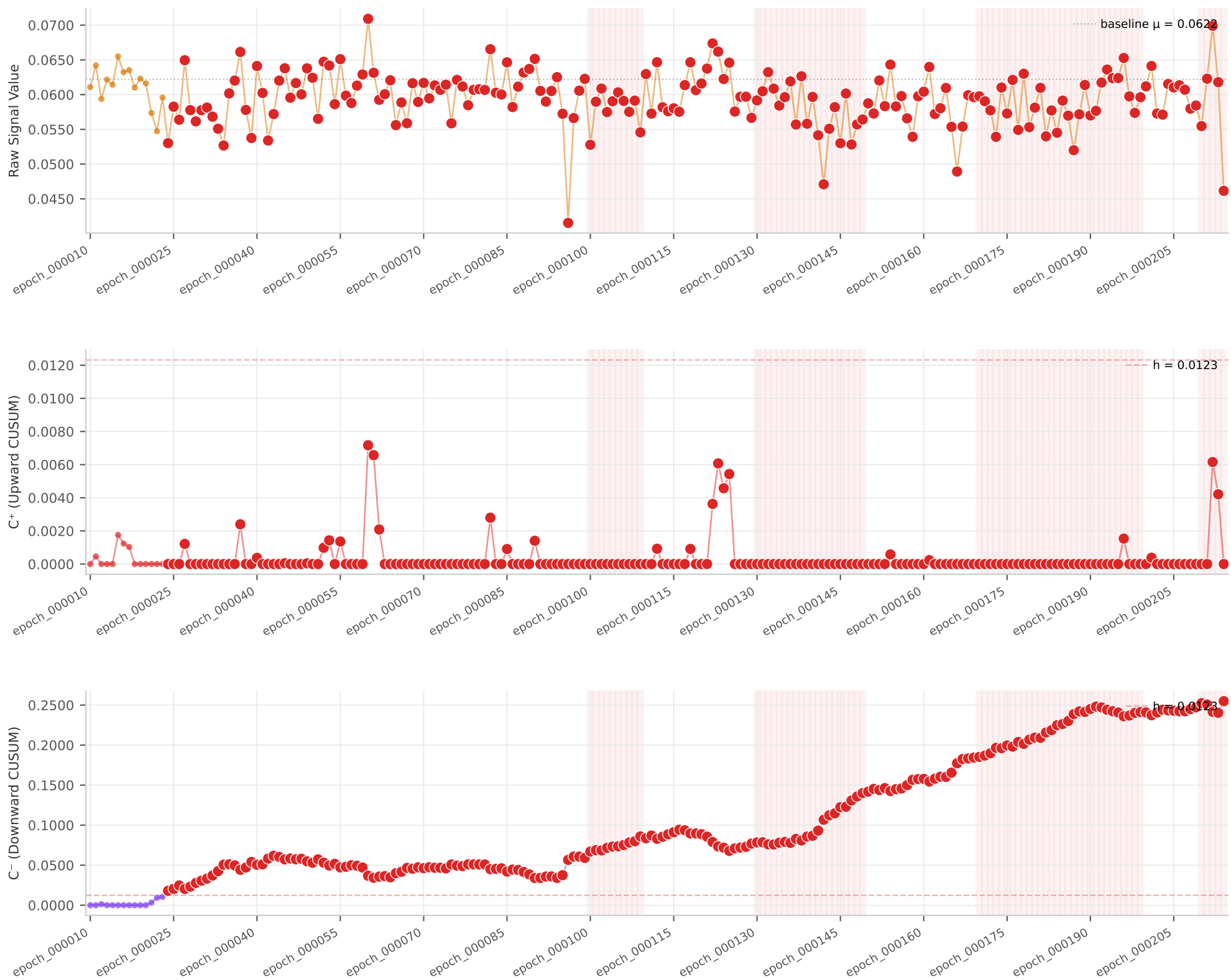
- CUSUM alarm fired
- Attacked epoch



Top-5 Source Concentration

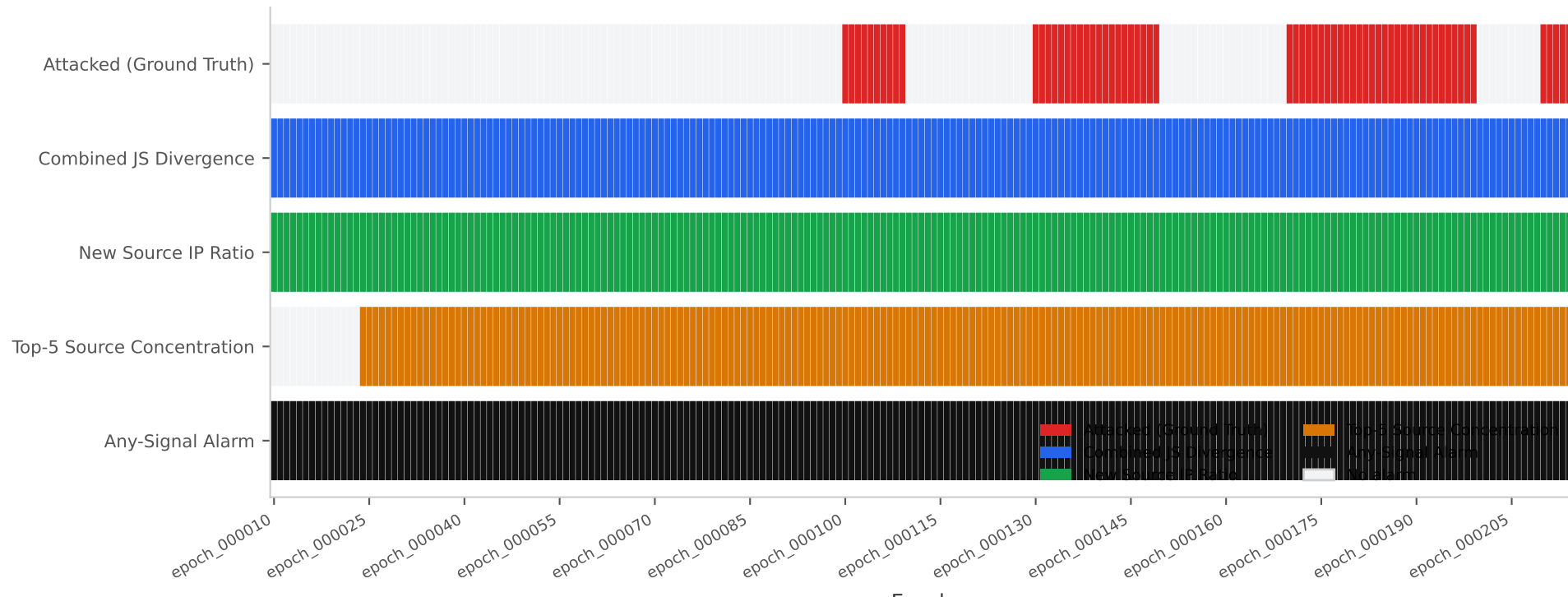
CUSUM Signal Analysis | malicious = 0% | baseline = 5%

- CUSUM alarm fired
- Attacked epoch



CUSUM Alarm Timeline

Filled cell = alarm fired for that epoch × signal



Detection Performance Metrics

Dataset: txtD280 | Malicious: ?% | Baseline: ?%

Metric	Value	Description
True Positives	65	Attacked epochs correctly alarmed
False Positives	140	Clean epochs incorrectly alarmed
True Negatives	0	Clean epochs correctly not alarmed
False Negatives	0	Attacked epochs missed
Precision	0.3171	$TP / (TP + FP)$
Recall	1.0000	$TP / (TP + FN)$
F1 Score	0.4815	Harmonic mean of Precision & Recall
Detection Rate	100.00%	Fraction of attacked epochs detected
False Alarm Rate	100.00%	Fraction of clean epochs with alarm